Lecture 21 (B20): Continuous-Time Heterogeneous Agents — Numerics

Solving CT Heterogeneous Agent Models: From Finite Differences to PINNs and EMINNs

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Based on: Achdou, Han, Lasry, Lions & Moll (2022), *Rev. Econ. Stud.* 89(1), 45–86.

Gu, Lauriere, Merkel & Payne (2024), "Global Solutions to Master Equations." arXiv:2406.13726.

Self-contained slides; Python implementations are introduced after the theory.

Materials: `https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models`

Recap and Goals

From Deck 1 (this morning): We derived the theory of CT heterogeneous agent models:

- Stochastic calculus: Brownian motion, Itô's lemma, Poisson jumps
- KFE (Fokker–Planck): density evolution, from physics to economics
- HJB: optimal consumption-savings policy
- Equilibrium: coupled HJB + KFE + market clearing (Huggett, Aiyagari, KS)
- Master equation: single PDE encoding the full equilibrium

This deck: How to solve these models numerically.

1. **Part I**: Traditional approach (finite differences) — brief overview, limitations
2. **Part II**: PINNs for the coupled HJB + KFE system (stationary: Huggett/Aiyagari)
3. **Part III**: EMINNs for the master equation (with aggregate shocks: Krusell–Smith)
4. **Part IV**: Results, comparison, and the "deep learning conundrum"

Table of Contents

Traditional Approach: Finite Differences

PINNs for HJB + KFE

EMINNs for the Master Equation

Results, Comparison, and Outlook

Part I

Traditional Approach: Finite Differences

The Benchmark We Want to Beat

Finite Differences: The Classical Approach

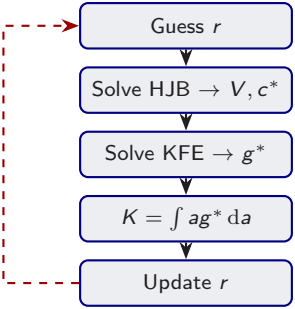
How have economists traditionally solved these models?

Finite difference method (Achdou et al., 2022):

1. Discretize wealth a on a grid $\{a_1, \ldots, a_I\}$
2. Approximate derivatives by finite differences:

$$V'(a_i) \approx \frac{V_{i+1} - V_i}{\Delta a} \quad \text{or} \quad \frac{V_i - V_{i-1}}{\Delta a}$$

3. Upwind scheme: use forward diff. when savings > 0, backward when < 0
4. Iterate HJB $\rightarrow$ policy $\rightarrow$ KFE $\rightarrow$ aggregates $\rightarrow$ prices $\rightarrow$ HJB



This summarizes the finite-difference workflow; code examples are provided later in Python.

Finite Differences: What They Produce

Finite-Difference Example: Aiyagari GE

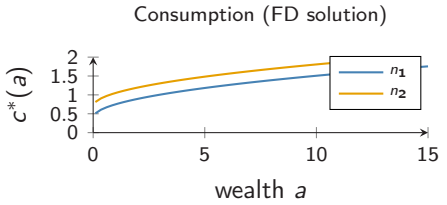
The classical FD approach solves:

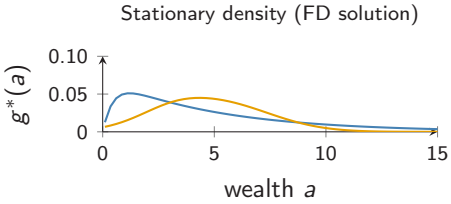
- HJB block: upwind discretization + implicit updates
- KFE block: transition operator + stationary distribution
- Price block: root-finding for market-clearing r

Produces:

- Value function $V(a, n)$
- Consumption/savings policy $c^*(a, n)$
- Stationary density $g^*(a, n)$
- Equilibrium prices r^*, w^*

This is our ground truth for validating deep





Why Finite Differences Are Not Enough

FD works well for:

- ✓ Stationary HJB–KFE systems
- ✓ Low-dimensional states (a, n)
- ✓ Monotone convergence theory
- ✓ Fast sparse linear algebra

FD is strained by:

- ✗ Global KS master equation
- ✗ State (a, n, z, g) with distribution g
- ✗ Dense tensor-product grids
- ✗ Moment truncation of g

Takeaway

Neural methods do not remove the equilibrium conditions; they approximate value and distribution maps without building a full grid over g.

Part II

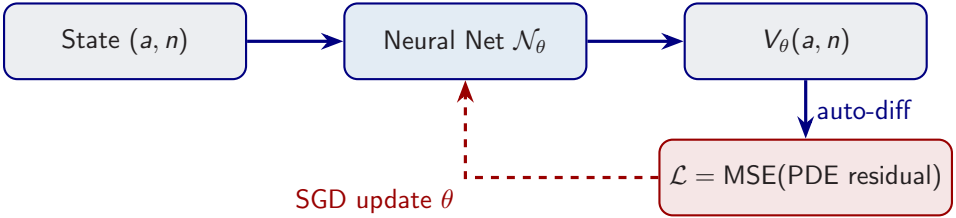
PINNs for the Coupled HJB + KFE System

Solving Huggett and Aiyagari with Neural Networks

Recap: What Is a PINN?

From Block 1 (earlier today): A Physics-Informed Neural Network (PINN) solves a PDE by:

1. Parameterizing the solution by a neural network: $u(x) \approx \mathcal{N}_\theta(x)$
2. Computing PDE residuals via automatic differentiation
3. Minimizing the residual over collocation points (no grid needed!)



Key advantage: mesh-free — no grid needed, scales to high dimensions.

Partial Equilibrium HJB: Stepping Stones

Before tackling full GE (Aiyagari), solve the HJB alone with exogenous prices.

Discrete Income (1D)

- 2-state Markov chain $z \in \{z_L, z_H\}$
- Coupled system: $V_1(a), V_2(a)$
- FOC eliminates max; upwind FD vs PINN
- **Notebook 01**

OU Diffusion (2D)

- Earnings: $dz = \theta(\mu - z)\,dt + \sigma\,dW$
- Single PDE: $V(a, z)$, second-order in z
- Neumann BCs; 2D auto-diff PINN
- **Notebook 02**

Key insight: PINNs scale gracefully from 1D→2D — no grid explosion.

PE Discrete Income: Model and HJB

Setup (exogenous prices r, w):

- CRRA utility $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, discount ρ
- Budget: $\dot{a} = r\,a + w\,z - c, \quad a \geq \underline{a}$
- Income: 2-state Markov chain
  $z \in \{z_L, z_H\}$
  with Poisson rates $\lambda_{12}, \lambda_{21}$

FOC eliminates max:

$$u'(c) = V_j'(a) \implies \boxed{c_j^* = (V_j')^{-1/\gamma}}$$

Coupled HJB for states $j = 1, 2$:

$$\boxed{\rho\, V_j = u(c_j^*) + V_j'\, s_j + \lambda_{jk}(V_k - V_j)}$$

where $s_j = r\,a + w\,z_j - c_j^*$ is savings.

Boundary conditions:

- $a = \underline{a}$: enforce $s_j = r\underline{a} + wz_j - c_j \geq 0$.
- If binding: $c_j = r\underline{a} + wz_j$ and boundary mass may form.
- $a = \bar{a}$: no outflow on truncated grid.

PE Discrete Income: Upwind Finite-Difference Scheme

Algorithm (implicit iteration)

1: Forward/backward FD of V:
$V_a^F = (V_{i+1} - V_i)/\Delta a, \;\; V_a^B = (V_i - V_{i-1})/\Delta a$

2: FOC inversion $\rightarrow$ consumption:
$c^F = (V_a^F)^{-1/\gamma}, \; c^B = (V_a^B)^{-1/\gamma}$

3: Savings candidates:
$s^F = \text{inc} - c^F, \;\; s^B = \text{inc} - c^B$

4: Upwind selection:
$\mathbb{I}^F = (s^F > 0), \;\; \mathbb{I}^B = (s^B < 0) \wedge \neg \mathbb{I}^F$

5: Build generator
$A = A_a(\text{upwind}) + \Lambda \otimes I_J$

6: Implicit step:
$\left(\frac{1}{\Delta} + \rho\right) V^{n+1} - A\, V^{n+1} = u(c) + \frac{V^n}{\Delta}$

Key design choices:

- Upwind: uses the derivative consistent with the drift direction $\rightarrow$ numerical stability
- Kronecker stacking: $\Lambda \otimes I_J$ couples the two income states into one $2J \times 2J$ system
- Implicit: unconditionally stable; large Δ accelerates convergence
- Converges in $\sim$13 iterations to $\|V^{n+1} - V^n\|_\infty < 10^{-8}$

Reference: Achdou et al. (2022, REStud), Appendix B.

PE Discrete Income: PINN Approach

Network architecture:

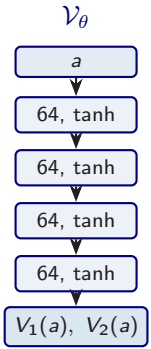
1 input (a) $\rightarrow$ 4$\times$64 `tanh` $\rightarrow$ 2 outputs (V_1, V_2)

Loss components:

Term	Description
$\mathcal{L}_{\text{PDE}}$	HJB residual MSE (both states)
$\mathcal{L}_{\text{BC}}$	Dirichlet at $\underline{a}, \bar{a}$ (from FD)
$\mathcal{L}_{\text{curv}}$	Concavity: $\text{relu}(V'')^2$

Key code (auto-diff):

```
V = net(a)                # [N, 2]
V_a = grad(V.sum(), a)    # V'
c = V_a ** (-1/gamma)     # FOC
s = income - c
R = rho*V - u(c) - V_a*s
    - lam*(V_swap - V)    # residual
```



No grid, no upwind logic — auto-diff handles all derivatives.

PE Diffusion Income: Model and 2D HJB

Earnings process (Ornstein–Uhlenbeck):

$$\boxed{dz = \theta(\mu - z)\, dt + \sigma\, dW}$$

with mean-reversion θ, long-run mean μ, volatility σ.

2D HJB (after FOC substitution):

$$\boxed{\rho\, V = u(c^*) + V_a\, s + \theta(\mu - z)\, V_z + \tfrac{1}{2}\sigma^2 V_{zz}}$$

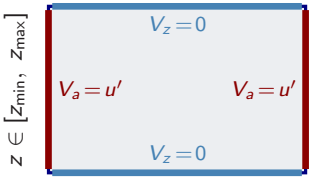
where $c^* = (V_a)^{-1/\gamma}$, $s = r\,a + w\,z - c^*$.

Key difference from 1D: the second-order diffusion term V_{zz}.

Domain & boundary conditions:

Left/right (a): Neumann $V_a = u'(\text{inc})$

Top/bottom (z): Reflecting $V_z = 0$



$a \in [\underline{a},\ \bar{a}]$

PE Diffusion: 2D Finite-Difference Scheme

Operator splitting by dimension:

Wealth direction (a): upwind as in 1D

- Forward/backward FD, FOC, upwind selection
- Produces sparse A_a (block-diagonal in z)

Earnings direction (z):

- Drift $\theta(\mu - z)$: upwind FD
- Diffusion $\frac{1}{2}\sigma^2 V_{zz}$: central FD
- Reflecting BCs: ghost-point symmetry ($V_{-1} = V_1$ at $z_{\min}$, $V_{J_z} = V_{J_z-2}$ at $z_{\max}$)
- Produces A_z of size $J_z \times J_z$

Combined generator:

$$A = A_a(s) + A_z \otimes I_{J_a}$$

Implicit step (same structure as 1D):

$$\left(\frac{1}{\Delta} + \rho\right) V^{n+1} - A\, V^{n+1} = u(c) + \frac{V^n}{\Delta}$$

Grid scaling:

	Classroom	Production
J_a	50	200
J_z	30	100
Unknowns	1,500	20,000

Performance:

- Converges in $\sim$9 implicit iterations
- Sparse direct solve (scipy `spsolve`)
- Cost grows as $O(J_a \times J_z)$ — manageable in 2D, but explodes in 3D+

PE Diffusion: 2D PINN

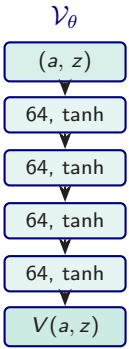
Network: 2 inputs $(a, z) \to 4\times64$ `tanh` $\to 1$ output V

Key ingredient — nested auto-diff for V_{zz}:

```
V     = net(a, z)
V_a   = grad(V, a)      # wealth
V_z   = grad(V, z)      # earnings
V_zz  = grad(V_z, z)    # diffusion!
```

Loss components (same structure as 1D):

Term	Description
$\mathcal{L}_{\text{PDE}}$	2D HJB residual MSE
$\mathcal{L}_{\text{BC}}$	Dirichlet on all 4 edges
$\mathcal{L}_{\text{curv}}$	Concavity: $\text{relu}(V_{aa})^2$



1D $\to$ **2D**: add z as 1 extra input neuron. No grid explosion — collocation cost is dimension-free.

From Partial to General Equilibrium

	PE Discrete (NB 01)	**PE Diffusion (NB 02)**	**GE Aiyagari (NB 03)**
PDEs solved	HJB only	HJB only	HJB + KFE
Prices (r, w)	exogenous	exogenous	endogenous
Income process	2-state Markov	OU diffusion	2-state Markov
State dimension	1D (a)	2D (a, z)	1D (a) + GE loop
PINN networks	V_θ only	V_θ only	$V_\theta + g_\phi$
FD unknowns	$2J$	$J_a \times J_z$	$2J$ + distribution

Progression of complexity:

1. **NB 01**: isolate the HJB with simplest income process $\rightarrow$ learn upwind FD and PINN mechanics
2. **NB 02**: extend to 2D with diffusion $\rightarrow$ mesh-free PINN advantage (V_{zz} via auto-diff)
3. **NB 03**: add the KFE for stationary distribution + market clearing $\rightarrow$ full Aiyagari GE

Notebook 03 adds the KFE and equilibrium conditions $\rightarrow$ full Aiyagari.

From Single PDE to Coupled System

Block 1 (PINNs): Solve a *single* PDE (Poisson, HJB cake-eating, Black–Scholes).
Block 2 (this block): Solve a *coupled system* of PDEs arising from equilibrium:

Component	**PDE**	**Unknown**
HJB (backward)	$\rho V = \max_c \{u(c) + V' \cdot s + \lambda(\Delta V)\}$	Value $V(a, n)$
KFE (forward)	$0 = -\partial_a[s \cdot g] - \lambda g + \lambda \hat{g}$	Density $g(a, n)$
Market clearing	$K = \sum_n \int a\, g \, \mathrm{d}a$	Prices r, w

The coupling: HJB output ($c^* \to s$) is the KFE input (drift).
KFE output ($g \to K, L$) determines prices (r, w) that enter HJB.

PINN approach: Parameterize *both* V and g by neural networks.
Minimize *all* residuals jointly via a single loss function.

Formal Target System (Stationary, Truncated Domain)

Train on $a \in [\underline{a}, \bar{a}]$, $n \in \{n_1, n_2\}$.

$$\rho V = \max_c \left\{ u(c) + V_a(wn + ra - c) + \lambda(n)\big(V(a, \hat{n}) - V(a, n)\big) \right\}, \quad c^* = (V_a)^{-1/\gamma}$$

$$0 = -\partial_a\big[s(a, n)g(a, n)\big] - \lambda(n)g(a, n) + \lambda(\hat{n})g(a, \hat{n}), \quad s = wn + ra - c^*$$

Boundary/state constraints:

$$V_a(\underline{a}, n) \geq u'(wn + r\underline{a}), \qquad s(\underline{a}, n) \geq 0$$

$$J(a, n) := s(a, n)g(a, n), \qquad J(\underline{a}, n) = 0$$

Market clearing:

$$\text{Huggett: } \sum_n \int_{\underline{b}}^{\bar{b}} b\, g(b, n)\, \mathrm{d}b = 0$$

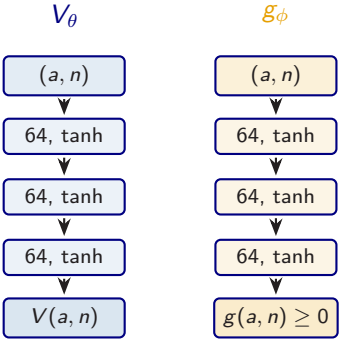
$$\text{Aiyagari: } K^s = \sum_n \int_{\underline{a}}^{\bar{a}} a\, g(a, n)\, \mathrm{d}a = K^d(r)$$

$$\sum_n \int_{\underline{a}}^{\bar{a}} g(a, n)\, \mathrm{d}a = 1$$

PINN Architecture for Aiyagari/Huggett

Two neural networks:

- $V_\theta(a, n)$: value function network
  - Input: (a, n) (n encoded as 0/1)
  - Output: scalar V
  - Architecture: 3–4 hidden layers, 64 neurons, tanh
- $g_\phi(a, n)$: density network
  - Input: (a, n)
  - Output: scalar $g \geq 0$ (use softplus)
  - Same architecture



Derived quantities (via auto-diff):

- $V'_\theta = \partial_a V_\theta \quad \Rightarrow$ consumption: $c^* = (V'_\theta)^{-1/\gamma}$
- $\partial_a g_\phi \quad \Rightarrow$ KFE residual

Explicit-Policy HJB PINN: $\mathcal{N}_\theta(a, n) \to (V, c)$

The cleaner formulation used in notebooks 06–07 makes the policy a network output, not a quantity hidden inside derivatives of V.

$$\mathcal{N}_\theta(a, n) \;=\; \big(\, V_\theta(a, n),\; c_\theta(a, n) \,\big), \qquad c_\theta = \mathrm{softplus}(\cdot) + \varepsilon \;>\; 0.$$

The loss enforces both the HJB residual and the FOC consistency:

$$\mathcal{L}^{\mathrm{HJB}} = \underbrace{\frac{1}{M}\sum_m R_{\mathrm{HJB}}\big(a_m, n_m;\; V_\theta, c_\theta\big)^2}_{\text{interior PDE residual}} + \kappa_{\mathrm{FOC}} \underbrace{\frac{1}{M}\sum_m \big(c_\theta(a_m, n_m)^{-\gamma} - V'_{\theta,a}(a_m, n_m)\big)^2}_{\text{first-order condition}}.$$

Why this is more teachable than $c = (V_a)^{-1/\gamma}$:

- The FOC is an explicit residual students can monitor.
- No early-training pathologies from $V_a \le 0$.
- Policy is a smooth network output, plots stable from epoch 1.
- FOC residual must vanish *independently* of the HJB residual.

Generalises to two states (nb. 06: V_1, V_2, c_1, c_2) and OU income (nb. 07: $V(a, z), c(a, z)$). Notebook 08 keeps $c = (V_a)^{-1/\gamma}$ and outputs (V, g) for the coupled HJB–KFE system.

PINN Objective: Interior + Boundary + Equilibrium

The total loss combines all equilibrium conditions:

$$\boxed{\mathcal{L} = \kappa_{\text{HJB}}\mathcal{L}^{\text{int}}_{\text{HJB}} + \kappa_{\text{KFE}}\mathcal{L}^{\text{int}}_{\text{KFE}} + \kappa_{\text{KFE,cum}}\mathcal{L}^{\text{cum}}_{\text{KFE}} + \kappa_{\text{SC}}\mathcal{L}_{\text{SC}} + \kappa_{\text{flux}}\mathcal{L}_{\text{flux}} + \kappa_{\text{mass}}\mathcal{L}_{\text{mass}} + \kappa_{\text{MC}}\mathcal{L}_{\text{MC}}}$$

Component	**Definition**	**What it enforces**
$\mathcal{L}^{\text{int}}_{\text{HJB}}$	$\frac{1}{M}\sum_m \lvert R_{\text{HJB}}(a_m, n_m)\rvert^2$	Interior HJB
$\mathcal{L}^{\text{int}}_{\text{KFE}}$	$\frac{1}{M}\sum_m \lvert R_{\text{KFE}}(a_m, n_m)\rvert^2$	Interior KFE
$\mathcal{L}^{\text{cum}}_{\text{KFE}}$	$\frac{1}{I}\sum_k \lvert R^{\text{cum}}_{\text{KFE}}(a_k)\rvert^2$	Integrated KFE balance
$\mathcal{L}_{\text{SC}}$	$\sum_n \max(0, -s_\theta(\underline{a}, n))^2$	State constraint
$\mathcal{L}_{\text{flux}}$	$\sum_n (J(\underline{a}, n)^2 + J(\bar{a}, n)^2)$	Boundary fluxes
$\mathcal{L}_{\text{mass}}$	$\left(\sum_n \int g_\phi \, \mathrm{d}a - 1\right)^2$	Normalization
$\mathcal{L}_{\text{MC}}$	Huggett: $(\int bg)^2$; Aiyagari: $(K^s - K^d(r))^2$	Market clearing

Softplus output enforces $g_\phi \geq 0$. Weights can be adapted with ReLoBRaLo.

HJB Residual: Computed via Auto-Differentiation

The interior HJB residual at collocation point (a, n). In notebook 08, consumption is recovered from the value derivative, $c^* = (V'_\theta)^{-1/\gamma}$. In notebooks 06–07, replace c^* by c_θ and add the explicit FOC residual from the previous slide.

$$R_{\text{HJB}}(a, n) = \rho V_\theta - u(c^*) - V'_\theta \cdot (wn + ra - c^*) - \lambda(n)\big(V_\theta(a, \hat{n}) - V_\theta(a, n)\big)$$

```
def hjb_residual(model_V, a, n, r, w, params):
    a.requires_grad_(True)
    V = model_V(a, n)
    V_a = torch.autograd.grad(V, a, ...)[0]   # auto-diff
    c_star = V_a.clamp(min=1e-8) ** (-1.0 / gamma)   # FOC
    s = w * n + r * a - c_star   # savings
    V_hat = model_V(a, n_hat)   # value in other state
    res = rho * V - u(c_star) - V_a * s
          - lam(n) * (V_hat - V)
    return res
```

Integrated KFE: A Stable Companion

Differentiating a noisy density g_ϕ pointwise is fragile; cumulative probability balance is often more stable.

Local (strong-form) residual.

$$R_{\mathsf{KFE}}(a,n) = \partial_a\big[s\, g_\phi\big] + \lambda(n) g_\phi(a,n) - \lambda(\hat{n}) g_\phi(a,\hat{n}).$$

Wiggles in g_ϕ produce large ∂_a noise.

Integrated flux balance on a deterministic grid $\{a_k\}$, with $J_j = s_j g_j$:

$$R^{\mathsf{cum}}_{1,k} = J_1(a_k) - J_1(\underline{a}) + \int_{\underline{a}}^{a_k} \big[\lambda_{12} g_1(u) - \lambda_{21} g_2(u)\big]\, \mathrm{d}u,$$

$$R^{\mathsf{cum}}_{2,k} = J_2(a_k) - J_2(\underline{a}) + \int_{\underline{a}}^{a_k} \big[\lambda_{21} g_2(u) - \lambda_{12} g_1(u)\big]\, \mathrm{d}u.$$

No $\partial_a g_\phi$; only quadrature.

- Cheap: one quadrature per epoch on a fixed grid.
- Improves stationary-density diagnostics in notebook 08.
- Use both: $\mathcal{L}^{\mathsf{KFE}} = \kappa_{\mathsf{loc}} \mathcal{L}_{\mathsf{loc}} + \kappa_{\mathsf{cum}} \mathcal{L}_{\mathsf{cum}}$.

KFE Residual: Density Must Satisfy the Forward Equation

The interior stationary KFE residual at (a, n):

$$R_{\mathrm{KFE}}(a, n) = \partial_a\big[s(a, n)\, g_\phi(a, n)\big] + \lambda(n)\, g_\phi(a, n) - \lambda(\hat{n})\, g_\phi(a, \hat{n})$$

```python
def kfe_residual(model_g, model_V, a, n, r, w, params):
    a.requires_grad_(True)
    g = model_g(a, n)
    c_star = get_consumption(model_V, a, n, params)
    s = w * n + r * a - c_star
    flux = s * g  # flux = savings * density
    d_flux = torch.autograd.grad(flux, a, ...)[0]
    g_hat = model_g(a, n_hat)  # density in other state
    res = d_flux + lam(n) * g - lam(n_hat) * g_hat
    return res
```

Boundary, Flux, and Market-Clearing Terms

Aiyagari market clearing:

$$\mathcal{L}_{\mathrm{MC}} = \left(\sum_n \int_{\underline{a}}^{\bar{a}} a\, g_\phi \,\mathrm{d}a - K^d(r)\right)^2$$

For Huggett replace with:

$$\mathcal{L}_{\mathrm{MC}}^{\mathrm{H}} = \left(\sum_n \int_{\underline{b}}^{\bar{b}} b\, g_\phi(b,n)\,\mathrm{d}b\right)^2 .$$

Normalization: $\mathcal{L}_{\mathrm{mass}} = \left(\sum_n \int_{\underline{a}}^{\bar{a}} g_\phi(a,n)\,\mathrm{d}a - 1\right)^2$

State-constraint penalty:

$$\mathcal{L}_{\mathrm{SC}} = \sum_n \max(0,\ -s_\theta(\underline{a},n))^2$$

Boundary flux penalty (truncated domain $[\underline{a},\bar{a}]$):

$$\mathcal{L}_{\mathrm{flux}} = \sum_n \left(J_\phi(\underline{a},n)^2 + J_\phi(\bar{a},n)^2\right), \qquad J_\phi(a,n) = s_\theta(a,n)\, g_\phi(a,n)$$

Integrals are evaluated via Monte Carlo or quadrature.

Training Algorithm: Jointly Solving HJB + KFE

PINN Training (Stationary Equilibrium)

```
Initialize V_θ, g_φ, guess r^(0)
Optional functional-form initialization for V_θ and g_φ; FD is reserved for validation
for k = 1, ..., N_epochs do
    Sample M collocation points {(a_m, n_m)}
    Compute c*_m = (V'_θ(a_m))^(-1/γ)
    Compute R^m_HJB, R^m_KFE via auto-diff
    Compute integrated KFE on a fixed asset grid
    Compute L_SC, L_flux, L_mass, L_MC
    Total loss: L = Σ κ_i L_i
    Update θ, φ via Adam
    if k mod N_r = 0 then
        Update r: r ← r + η(K^s − K^d)
    end if
end for
```

Key choices:

- **Collocation**: sample (a, n) uniformly on $[\underline{a}, \bar{a}] \times \{n_1, n_2\}$
- **Price update**: update r slowly (every N_r steps) to allow NN to adapt
- **Loss weights**: start with $\kappa_{\text{HJB}} = \kappa_{\text{KFE}} = 1$, larger $\kappa_{\text{SC}}, \kappa_{\text{flux}}, \kappa_{\text{MC}}$
- **Optimizer**: Adam with lr $\sim 10^{-3}$, then deterministic FP64 L-BFGS
- **Validation**: compare against FD only after training

Same mesh-free philosophy:
economic equations = loss function
auto-diff = all derivatives

Applying the PINN: Huggett (1993) Model

Huggett model (endowment economy, zero net supply of bonds):

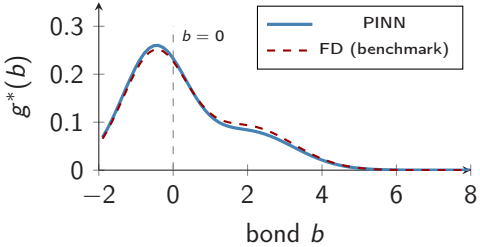
Differences from Aiyagari:

- Asset: risk-free bond b (not capital)
- Budget: $db = (qb + n - c)\,\mathrm{d}t$
- Market clearing: $\sum_n \int b\, g(b, n)\,\mathrm{d}b = 0$
- Price: bond return q (not interest rate via firm FOC)

PINN modification:

- Same architecture (V_θ, g_ϕ)
- HJB: replace ra by qb, remove wage
- $\mathcal{L}_{\mathrm{MC}} = \left(\int b\, g_\phi\,\mathrm{d}b\right)^2$
- Iterate on q instead of r

Huggett: PINN vs. FD



PINN closely matches the FD benchmark.

Applying the PINN: Aiyagari (1994) Model

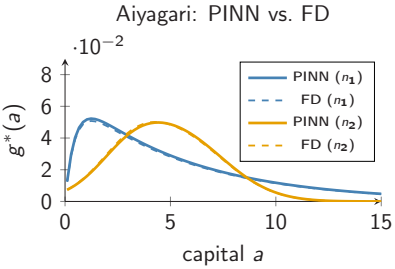
Aiyagari model (production economy, capital market clearing):

vs. Huggett: two prices (r, w) from firm FOC, $\int ag = K$, $\int ng = L$, $r = \alpha K^{\alpha-1} L^{1-\alpha} - \delta$.

PINN diagnostics (notebook 08):

Diagnostic	Value
K^{FD} / K^{PINN}	0.30893 / 0.30894
$\lvert K^{\text{FD}} - K^{\text{PINN}} \rvert$	8.7×10^{-6}
c MSE	4.7×10^{-4}
g MSE	8.1×10^{-4}
KS dist. z_1/z_2	0.066/0.059
$\max \lvert R_{\text{HJB}} \rvert$	5.7×10^{-2}
$\max \lvert R_{\text{KFE}} \rvert$	2.7×10^{-1}

Honest read: K matches to 5 digits; pointwise density and KS gaps are non-trivial. The integrated-KFE term targets exactly the loosest residual.



Right-skewed, mass at $\underline{a}$: reproduced by PINN.

Handling the Borrowing Constraint

The borrowing constraint $a \geq \underline{a}$ is a key challenge for PINNs.

Option A: Soft constraint (penalty)

Add penalty: $\psi(a) = -\frac{1}{2}\kappa(a - \underline{a})^2$ for $a \leq \underline{a}$

Advantages:

- Simple to implement
- Smooth loss landscape
- Avoids mass point issues in KFE

Option B: Hard constraint (trial solution)

Architecture: $V_\theta(a, n) = (a - \underline{a})\hat{V}_\theta + V_{\text{BC}}$

Advantages:

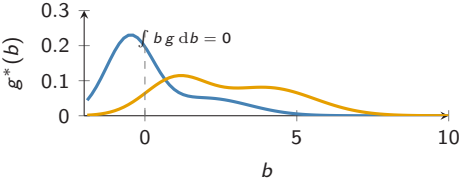
- Exact BC satisfaction
- No tuning of κ
- Recall the PINN warm-up notebook

Recommendation (Achdou et al., 2022; Gu et al., 2024):

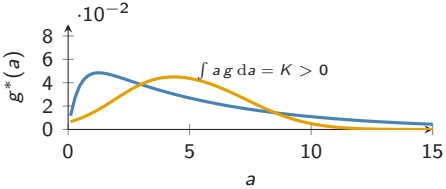
Use the soft borrowing constraint $\psi(a)$. The inequality BC causes technical problems (non-differentiable V, mass point in g). Softening avoids these while being economically faithful for κ large enough.

PINN Solutions: Comparing Huggett and Aiyagari Distributions

Huggett (bond economy)



Aiyagari (production economy)



- Mean wealth $= 0$ (zero net supply)
- Large mass at $\underline{b}$ (constrained)
- $q^* < \rho$ (low return)

- Mean wealth $= K > 0$ (capital stock)
- Long right tail (wealthy agents)
- $r^* < \rho$ but capital earns return

The PINN correctly captures these qualitative differences. Same code architecture, different market clearing conditions.

Why PINNs over Finite Differences?

For stationary problems (Huggett/Aiyagari without aggregate shocks):

	Finite Differences	**PINN**
Grid required	Yes (I points)	No (collocation)
Upwind scheme	Manual implementation	Auto-diff handles it
Curse of dimensionality	$O(I^d)$	$O(M \cdot \lvert\theta\rvert)$
Extension to higher dim	Very hard	Natural
Borrowing constraint	Special treatment	Soft penalty / hard BC
Accuracy (2D)	$\sim 10^{-6}$	$\sim 10^{-4}$
Speed (2D)	Faster	Slower

Verdict: ***FD is faster and more accurate*** for low-dim stationary problems — use it as the gold-standard benchmark. PINNs scale to high dimensions: aggregate shocks (KS, state $\ni g$), multiple assets, richer heterogeneity.
$\Longrightarrow$ Part III: **EMINNs** for the master equation.

Part III

EMINNs: Solving the Master Equation

Economic Model Informed Neural Networks

Gu, Lauriere, Merkel & Payne (2024)

Note: the companion EMINN notebook is deferred to the next iteration of the course; this Part III is conceptual.

The Challenge: Aggregate Shocks

Krusell–Smith (1998): Add aggregate TFP shocks $dZ_t = \eta(\bar{Z} - Z_t)\,\mathrm{d}t + \sigma^z\,\mathrm{d}B_t^0$.
Now the value function depends on the entire distribution:

$$V = V(a, n, z, g) \qquad \text{where } g \text{ is infinite-dimensional}$$

Why is this hard?

- Cannot discretize g on a grid (infinite-dimensional)
- Original KS trick: $g \approx K$ (first moment). But loses distributional information.
- We want global solutions: accurate for all (z, g), not just near steady state

The Master Equation (from Deck 1)

Substitute KFE + market clearing + belief consistency into HJB $\Rightarrow$ single PDE:

$$\mathcal{L}V(a, n, z, g) = 0$$

One PDE to solve, but with an infinite-dimensional argument g.

Approximating the Distribution: Three Approaches

Key idea: Replace the infinite-dimensional g by a finite-dimensional approximation $\hat{\varphi}$.

	Finite Population	**Discrete State**	**Projection**
Dist. params $\hat{\varphi}_t$	$\{(a_t^i, n_t^i)\}_{i \leq N}$	Masses on grid	Basis coefficients
Dist. approx. $\hat{g}_t$	$\frac{1}{N}\sum \delta_{\hat{\varphi}_t^i}$	$\sum \hat{\varphi}_{i,j,t}\delta_{(a_i,n_j)}$	$b_0 + \sum \hat{\varphi}_{i,t} b_i(a,n)$
KFE approx.	Other agents' evolution	FD mass evolution	Coeff. evolution
Dimension	~ 40	~ 200	~ 5
$K_t =$	$\frac{1}{N}\sum a_t^i$	$\sum_j a_j \hat{\varphi}_{j,t}$	$\int a \sum \hat{\varphi}_{i,t} b_i \, \mathrm{d}a$

After approximation: $V(a,n,z,g) \approx \hat{V}(a,n,z,\hat{\varphi})$ where $\hat{\varphi} \in \mathbb{R}^N$.
Now the master equation becomes a finite-dimensional PDE that a neural network can solve!

Approach 1: Finite Population ($N \approx 40$ agents)

Replace continuum by N agents with states $\hat{\varphi}_t = \{(a_t^1, l_t^1), \ldots, (a_t^N, l_t^N)\}$.

Approximate capital: $K_t = \frac{1}{N} \sum_{i=1}^{N} a_t^i$

Why are 40 agents sufficient?

- We sample individual states (a, n) and distribution states $(\hat{\varphi})$ separately when generating training data
- For individual policy: can concentrate on high-curvature regions (near $\underline{a}$) independently of N
- For aggregate capital: LLN kicks in quickly — 40 agents give accurate K
- Can analytically eliminate finite-agent noise in $\hat{\varphi}$-drift during training

This is much smaller than discrete-time simulation methods (e.g., Young's method (Lecture 15) used $\sim$ 10,000 agents). Continuous-time formulation helps!

Approach 2: Discrete State Space ($\sim$ 200 grid points)

Discretize wealth on a grid $\{a_1, \ldots, a_I\}$. Distribution $\approx$ masses on grid:

$$\hat{g}_t(a, n) \approx \sum_{i=1}^{I} \sum_{j=1}^{2} \hat{\varphi}_{i,j,t} \, \delta_{(a_i, n_j)}$$

KFE becomes: evolution of grid masses via finite differences.

Master equation: functional derivative w.r.t. g becomes a partial derivative w.r.t. each mass $\hat{\varphi}_{m,j}$:

$$\hat{\mathcal{L}}_g \hat{W} = \sum_{j=1,2} \sum_{m=1}^{I} \mu_{\hat{\varphi},m,j}(z, \hat{\varphi}) \, \partial_{\hat{\varphi}_{m,j}} \hat{W}$$

Dimension: $\hat{\varphi} \in \mathbb{R}^{2I}$ with $I \approx 100$, so $\dim(\hat{\varphi}) \approx 200$.

Neural network: needs an embedding to handle the 200-dim input efficiently.

$\Rightarrow$ Use a recurrent or set-encoder architecture (Sirignano & Spiliopoulos, 2018).

Approach 3: Projection onto Basis Functions (~ 5 components)

Idea: Project g onto a low-dimensional basis: $\hat{g}_t(a,n) = b_0(a,n) + \sum_{i=1}^{N} \hat{\varphi}_{i,t}\, b_i(a,n)$

Choice of basis (Gu et al., 2024):

- Use eigenfunctions of the steady-state KFE operator $\bar{\mathcal{L}}^{KF}$
- These are the most persistent components of the density
- Intuition: most persistent components carry most price-relevant information
- Only $N \approx 5$ basis functions needed!

Dimension: $\hat{\varphi} \in \mathbb{R}^5$. Total NN input: $(a, n, z, \hat{\varphi}) \in \mathbb{R}^{2+1+5} = \mathbb{R}^8$.

Projection is lowest-dimensional but requires the most intricate setup (computing eigenfunctions, choosing test functions for KFE evolution).

Comparing the Three Approximation Approaches

	Finite Population	**Discrete State**	**Projection**
Dimension of $\hat{\varphi}$	~ 40	~ 200	~ 5
NN architecture	Feed-forward	Recurrent + embedding	Feed-forward
Initialization	$W(a, \cdot) = e^{-a}$	Random	Random
Customization	Choose N	Grid, FD method	Basis, test functions
Conceptual difficulty	Low	Medium	High

Trade-off:

- Finite population: simple concept, moderate dimension, works well in practice
- Discrete state: most similar to FD, highest dimension, needs embedding
- Projection: lowest dimension, but requires careful choice of basis and test functions

All three approaches are validated in Gu et al. (2024). They produce consistent results.

Neural Network for $W := \partial_a V$

Parameterize $W(\omega) \approx \hat{W}_\Theta(\omega)$ where $\omega = (a, n, z, \hat{\varphi})$:

$$\boxed{\text{Input: } \omega = (a, n, z, \hat{\varphi}) \in \mathbb{R}^d}$$

$$\boxed{\text{Hidden 1: } \boldsymbol{h}^{(1)} = \phi^{(1)}(W^{(1)}\omega + b^{(1)}), \quad 64, \text{ tanh}}$$

$$\boxed{\text{Hidden 2: } \boldsymbol{h}^{(2)} = \phi^{(2)}(W^{(2)}\boldsymbol{h}^{(1)} + b^{(2)}), \quad 64, \text{ tanh}}$$

$$\vdots$$

$$\boxed{\text{Hidden 5: } \boldsymbol{h}^{(5)} = \phi^{(5)}(W^{(5)}\boldsymbol{h}^{(4)} + b^{(5)}), \quad 64, \text{ tanh}}$$

$$\boxed{\text{Output: } \hat{W} = \sigma(W^{(6)}\boldsymbol{h}^{(5)} + b^{(6)}), \quad \text{softplus}}$$

- $H = 5$ hidden layers, 64 neurons each, tanh activation
- Output: softplus $\sigma(x) = \log(1 + e^x)$ ensures $W > 0$ (marginal value positive)
- Parameters: $\Theta = (W^{(1)}, b^{(1)}, \ldots, W^{(6)}, b^{(6)})$
- Optimal consumption: $c^* = W^{-1/\gamma}$ (directly from the envelope condition)

The EMINN Algorithm

Economic Model Informed Neural Network (EMINN)

Initialize NN $\hat{W}_\Theta$ with parameters Θ
while Loss $>$ tolerance **do**
  Sample M points: $\boldsymbol{S} = (a_m, n_m, z_m, \hat{\varphi}_m)_{m=1}^M$
  Compute loss: $\mathcal{E} = \kappa^e \frac{1}{M} \sum_m |\hat{\mathcal{L}}(\cdot;\Theta)|^2 + \kappa^s \mathcal{E}^s(\boldsymbol{S};\Theta)$
    $\hat{\mathcal{L}}$: master eq. residual (automatic differentiation); $\mathcal{E}^s$: shape penalty
  Update $\Theta \leftarrow \Theta - \alpha \nabla_\Theta \mathcal{E}$
end while

This is exactly a PINN applied to the master equation!

- "Physics" = economic equilibrium conditions (master equation)
- All derivatives (including w.r.t. $\hat{\varphi}$) computed by automatic differentiation

Computing the Master Equation Residual

The residual $\hat{\mathcal{L}}$ decomposes into three operators:

$$0 = (r(z, \hat{\varphi}) - \rho)\,\hat{W} + \underbrace{\hat{\mathcal{L}}_x \hat{W}}_{\text{individual}} + \underbrace{\hat{\mathcal{L}}_z \hat{W}}_{\text{TFP}} + \underbrace{\hat{\mathcal{L}}_g \hat{W}}_{\text{distribution}}$$

Operator	**Expression**
$\hat{\mathcal{L}}_x \hat{W}$	$s(\cdot)\partial_a \hat{W} + \lambda(n)\big(\hat{W}(\hat{n}) - \hat{W}(n)\big)$ (savings + income switch)
$\hat{\mathcal{L}}_z \hat{W}$	$\partial_z \hat{W} \cdot \mu^z(z) + \frac{1}{2}(\sigma^z)^2 \partial_{zz} \hat{W}$ (OU aggregate shock)
$\hat{\mathcal{L}}_g \hat{W}$	$\sum_n \frac{\partial \hat{W}}{\partial \hat{\varphi}_n} \cdot \mu_{\hat{\varphi},n}(z, \hat{\varphi})$ (distribution evolution)

All derivatives $(\partial_a, \partial_z, \partial_{zz}, \partial_{\hat{\varphi}_n})$ computed by one backward pass through the neural network. This is the power of automatic differentiation: even the functional derivative (w.r.t. distribution) is just a partial derivative in the finite-dimensional approximation.

Shape Constraints: Avoiding "Cheat Solutions"

The Macroeconomic Deep Learning Conundrum

The algorithm is straightforward to describe and code... but hard to implement successfully! NNs may get stuck in bad solutions (constant, non-increasing, non-concave V).

Solution: Add shape penalties $\mathcal{E}^s$ to the loss:

- **Concavity**: penalize $\partial_{aa} V > 0$: $\mathcal{E}^{\text{concav}} = \frac{1}{M} \sum \max(0, \partial_{aa} V_m)^2$
- **Monotonicity**: penalize $\partial_{za} V > 0$ (higher $z \Rightarrow$ lower marginal value)

Additional tricks:

- Initialization: $W(a, \cdot) = e^{-a}$ (correct qualitative shape from the start)
- Architecture: train $\phi(a, n, z, g; \theta) \cdot (a - \underline{a})^{1-\gamma}$ (correct asymptotics)
- Active sampling: sample more points where the residual is large

Sampling Strategies: Where to Evaluate the Residual

Challenge: state space $(a, n, z, \hat{\varphi})$ is high-dimensional. Must sample from economically relevant regions.

Distribution $\hat{\varphi}$ sampling:

1. Moment: draw K, then $\hat{\varphi}$ consistent with K
2. Mixed steady-state: solve SS for various z, take mixtures
3. Ergodic: simulate economy with current $\hat{W}$, use simulated distributions

Individual states (a, n):

- Uniform on $[\underline{a}, \bar{a}] \times \{l_1, l_2\}$
- Add draws where error is high (active sampling)

Aggregate state z: Uniform on $[z_{\min}, z_{\max}]$

Separation of individual and distribution sampling is key to efficiency.

Implementation Details

	Finite Pop.	Discrete State	Projection
Neural Network			
Structure	Feed-forward	RNN/embed.	Feed-forward
Initialization	$W(a,\cdot) = e^{-a}$	Random	Random
Sampling			
(a, n)	Active	Uniform	Uniform
$\hat{\varphi}$	Moment sampling	Mixed steady-state	Ergodic
z	$U[z_{\min}, z_{\max}]$	$U[z_{\min}, z_{\max}]$	$U[z_{\min}, z_{\max}]$
Loss Function			
Shape constraints	$\partial_{aa} V < 0$, $\partial_{za} V < 0$	Same	$\partial_{aa} V < 0$
Weights	$\kappa^e = 100$, $\kappa^s = 1$	$\kappa^e = \kappa^s = 1$	$\kappa^e = \kappa^s = 1$

Q&A: Key differences from discrete time?

- Derivatives replace expectations, so automatic differentiation is essential.
- Collocation and shape constraints replace simulation grids and false-time steps.

Part IV

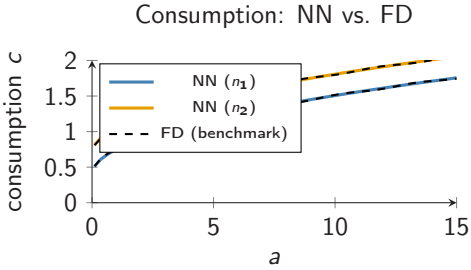
Results, Comparison, and Outlook

Results: Aiyagari Model (No Aggregate Shocks)

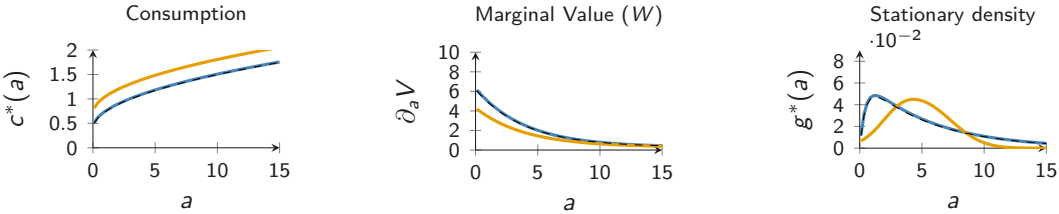
Validation: Compare EMINN solutions against finite-difference benchmark.

Method	ME Loss	MSE(NN,FD)
Finite Pop. NN	3.1×10^{-5}	4.8×10^{-5}
Discrete State NN	9.3×10^{-6}	6.6×10^{-5}

- ME residual: $\sim 10^{-5}$ (excellent)
- Agreement with FD: $\sim 10^{-5}$ MSE
- Both approaches match FD closely
- Consistent across multiple runs



Aiyagari: Consumption, Marginal Value, and Distribution



Solid curves: NN Dashed curves: FD benchmark

NN and FD are very close across all three panels.

Reported Results: Krusell–Smith with Aggregate Shocks

Gu et al. (2024) report global master-equation solutions with aggregate TFP shocks.

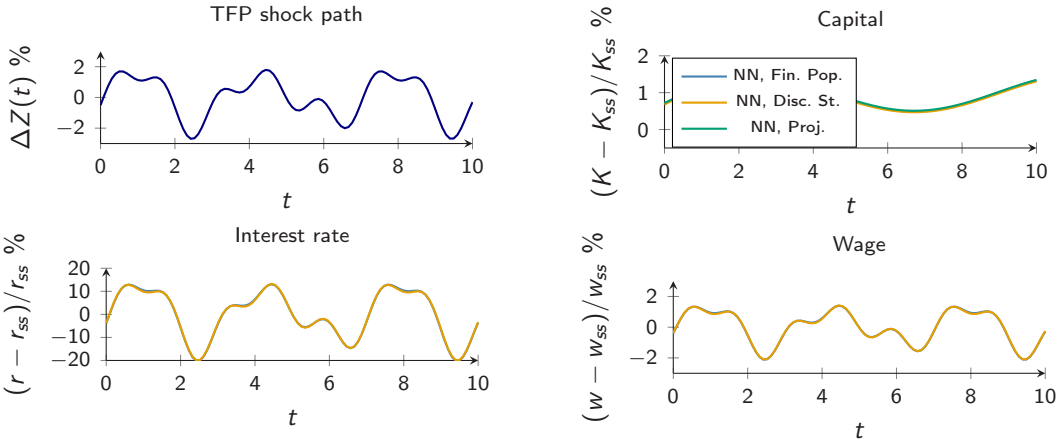
Method	Master Eq. Training Loss
Finite Pop. NN	3.0×10^{-5}
Discrete Grid NN	9.6×10^{-5}
Projection NN	8.5×10^{-6}

All three approaches achieve reported training losses around 10^{-5}.

Comparison with Fernandez-Villaverde, Hurtado & Nuno (2023):

- Their method: perturbation around steady state
- EMINN: neural global approximation of the master equation
- Reported paths match closely for moderate shocks; differences matter farther from steady state

Krusell–Smith: Sample Time Paths



All three NN approaches produce nearly identical time paths for aggregate variables.

Method Comparison: FD vs. PINN vs. EMINN

	Finite Diff.	**PINN**	**EMINN**
Global master eq.	✗	✗	✓
Stationary equilibrium	✓	✓	✓
Grid required	Yes	No	No
High-dimensional scaling	Limited	Better, validate	Better, validate
Handles g as state	Standard: no	No	Via $\hat{\varphi}$
Theory	Strong in low dim	Limited	Limited
Low-dim accuracy	$\sim 10^{-6}$	problem-dependent	reported $\sim 10^{-5}$

Bottom line:

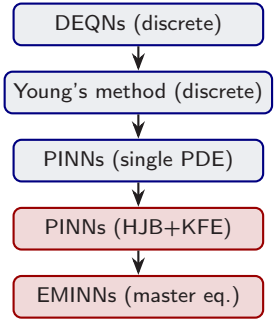
- **Stationary, low-dim**: FD is fast and reliable. Use it as benchmark.
- **Aggregate shocks**: EMINNs target global master-equation solutions.
- **PINNs**: useful bridge for the coupled HJB–KFE system.

The Deep Learning Advantage

Why deep learning for het. agent macro?

1. Meshless: no exponential grid growth
2. Handles ∞-dim states: distribution g approximated by finite $\hat{\varphi}$, NN handles the rest
3. Auto-diff: all derivatives (incl. functional) computed automatically
4. Global: accurate for all states, not just near steady state
5. Flexible: same framework for Huggett, Aiyagari, KS, HANK, ...

Connection to this course:



Open Challenges

The Macroeconomic Deep Learning Conundrum (Gu et al., 2024)

"The algorithm is straightforward to describe and code. . . but hard to implement successfully!"

Key challenges:

1. **Training stability**: NN can get stuck in bad local minima ("cheat solutions")
   - Constant, non-increasing, or non-concave $V \Rightarrow$ shape penalties needed
2. **No convergence guarantees**: unlike FD, we cannot prove the NN converges
3. **Sensitivity to hyperparameters**: architecture, lr, sampling, loss weights all matter
4. **Verification**: must compare against known solutions (FD, perturbation)
   - First validate on Aiyagari (no agg. shocks), then extend to KS
5. **Computational cost**: training can take hours/days on GPU for large models

Active research area: better architectures, better sampling, better loss functions.

Extensions and Future Directions

The EMINN framework extends naturally to richer models:

Model extensions:

- **HANK**: add nominal rigidities (Kaplan, Moll & Violante, 2018)
- **Multi-asset**: bonds + stocks + housing (higher-dim individual state)
- **Life-cycle**: age as additional state (more heterogeneity)
- **Non-convex adjustment costs** (e.g., lumpy investment)

Methodological extensions:

- **Better sampling**: importance sampling, adaptive collocation
- **Transfer learning**: pre-train on simpler model, fine-tune on complex one
- **Multi-scale training**: curriculum from easy $\rightarrow$ hard
- **Hybrid FD-NN**: use FD for low-dim parts, NN for high-dim parts

The frontier: Can we solve HANK models with aggregate shocks globally? This is one of the most important open problems in computational macro.

Summary and Key Takeaways

FD (Benchmark)

Fast and accurate for stationary, low-dim problems. Gold standard for Aiyagari/Huggett.

PINN (Coupled PDEs)

Mesh-free solution of HJB + KFE via auto-diff. Same mesh-free philosophy.

EMINN (Master Eq.)

Global solutions with aggregate shocks. Handles distribution as state.

The big picture:

1. CT het. agent models = coupled HJB + KFE + market clearing
2. Deep learning (PINNs/EMINNs) embeds equilibrium conditions in the loss
3. Auto-differentiation computes all derivatives (including functional derivatives)
4. For aggregate shocks: approximate g in finite dimensions, solve master equation
5. Validated against FD benchmarks — works, but training requires care

Next block: Surrogate models & Gaussian Processes.

Code Companion: Run This Live

Companion notebooks (self-contained, PyTorch):

- **NB 06** *(self-study)*: `06_PE_Discrete_HJB_PINN.ipynb` — 1D PE with discrete income: upwind FD + PINN
- **NB 07** *(self-study)*: `07_PE_Diffusion_HJB_PINN.ipynb` — 2D PE with OU earnings: 2D FD + PINN
- **NB 08** *(in-class)*: `08_Aiyagari_Continuous_Time_FD_and_PINN_PyTorch.ipynb` — full GE Aiyagari: HJB + KFE + market clearing

Suggested in-class demo flow:

1. Start with NB 06 (simplest, self-study): FD convergence, then PINN overlay
2. Move to NB 07 (self-study): 2D contour plots, cross-sections, mesh-free advantage
3. Focus on NB 08 (in-class): full GE loop, PINN vs FD agreement diagnostics

References

Core papers:

- Achdou, Y., Han, J., Lasry, J.-M., Lions, P.-L. & Moll, B. (2022). Income and Wealth Distribution in Macroeconomics: A CT Approach. *REStud* 89(1), 45–86.
- Gu, Z., Lauriere, M., Merkel, S. & Payne, J. (2024). Global Solutions to Master Equations for CT HA Models. arXiv:2406.13726.
- Fernandez-Villaverde, J., Hurtado, S. & Nuno, G. (2023). Financial Frictions and the Wealth Distribution. *Econometrica* 91(3), 869–901.

Foundational:

- Aiyagari, S.R. (1994). Uninsured Idiosyncratic Risk and Aggregate Saving. *QJE* 109(3), 659–684.
- Huggett, M. (1993). The Risk-Free Rate in HA Economies. *JEDC* 17(5–6), 953–969.
- Krusell, P. & Smith, A.A. (1998). Income and Wealth Heterogeneity. *JPE* 106(5), 867–896.

Mean field games and DL:

- Lasry, J.-M. & Lions, P.-L. (2007). Mean Field Games. *Jpn. J. Math.* 2(1), 229–260.
- Sirignano, J. & Spiliopoulos, K. (2018). DGM: A Deep Learning Algorithm for Solving PDEs. *J. Comput. Phys.* 375, 1339–1364.

HANK and applications:

- Kaplan, G., Moll, B. & Violante, G.L. (2018). Monetary Policy According to HANK. *AER* 108(3), 697–743.
- Ahn, S., Kaplan, G., Moll, B., Winberry, T. & Wolf, C. (2018). When Inequality Matters for Macro. *NBER Macro Annual* 32(1), 1–75.
- Gabaix, X., Lasry, J.-M., Lions, P.-L. & Moll, B. (2016). The Dynamics of Inequality. *Ecta* 84(6), 2071–2111.
- Nuno, G. & Moll, B. (2018). Social Optima in Economies with HA. *RED* 28, 150–180.

Moll lecture notes:

- benjaminmoll.com/lectures/ and benjaminmoll.com/codes/

Questions?

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https://sischei.github.io

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Covered today (Deck 2):

I FD: The Benchmark
II PINNs for HJB+KFE
III EMINNs for Master Eq.
IV Results & Outlook

Next block:

Surrogate Models & Gaussian Processes

09:00–12:00